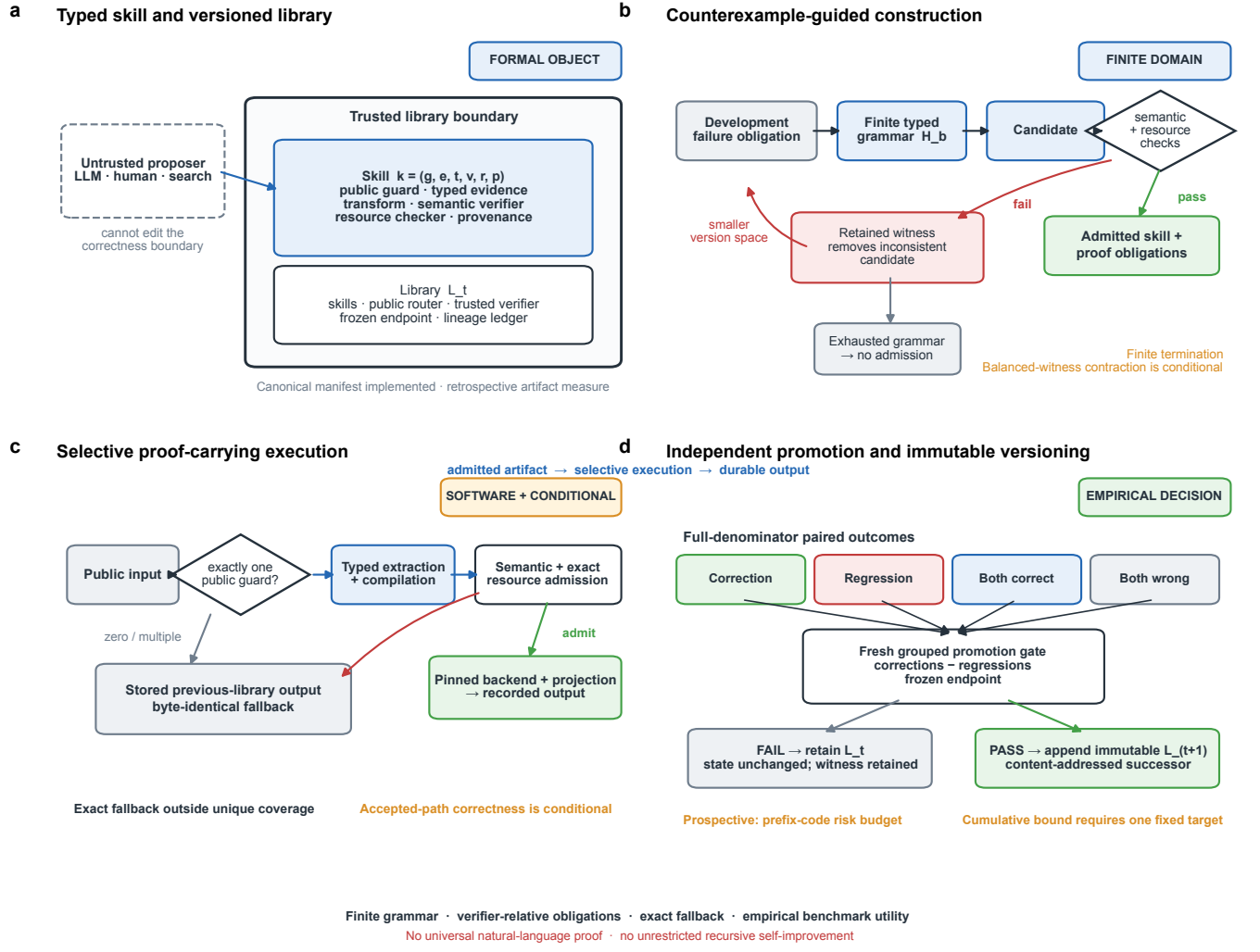
a
Typed skill and versioned library
FORMAL OBJECT
Trusted library boundary
Untrusted proposer
LLM · human · search
cannot edit the
correctness boundary
Skill k = (g, e, t, v, r, p)
public guard · typed evidence
transform · semantic verifier
resource checker · provenance
Library L_t
skills · public router · trusted verifier
frozen endpoint · lineage ledger
Canonical manifest implemented · retrospective artifact measure
b
Counterexample-guided construction
FINITE DOMAIN
Development
failure obligation
Finite typed
grammar H_b
Candidate
semantic
+ resource
checks
fail
pass
smaller
version space
Retained witness
removes inconsistent
candidate
Admitted skill +
proof obligations
Exhausted grammar
→ no admission
Finite termination
Balanced-witness contraction is conditional
c
Selective proof-carrying execution
admitted artifact → selective execution → durable output
SOFTWARE + CONDITIONAL
Public input
exactly one
public guard?
Typed extraction
+ compilation
Semantic + exact
resource admission
zero / multiple
admit
Stored previous-library output
byte-identical fallback
Pinned backend + projection
→ recorded output
Exact fallback outside unique coverage
Accepted-path correctness is conditional
d
Independent promotion and immutable versioning
EMPIRICAL DECISION
Full-denominator paired outcomes
Correction
Regression
Both correct
Both wrong
Fresh grouped promotion gate
corrections − regressions
frozen endpoint
FAIL → retain L_t
state unchanged; witness retained
PASS → append immutable L_(t+1)
content-addressed successor
Prospective: prefix-code risk budget
Cumulative bound requires one fixed target
Finite grammar · verifier-relative obligations · exact fallback · empirical benchmark utility
No universal natural-language proof · no unrestricted recursive self-improvement